Assignment 1 - Git & GitHub Fundamentals

Session 1 - Group A | Individual Lab Report

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Course Code & Section: CSC 3324 03

GitHub Username: N00rFaR

Repository: https://github.com/N00rFaR/git-session1-N00rFaR

1. Repository Setup

I renamed the supplied STARTER project to git-session1-N00rFaR and initialized a Git repository with main as its initial branch. My .gitignore contains local.env, *.log, and !important.log. These rules exclude the local configuration and ordinary logs while allowing important.log to be tracked. I verified the repository root with git rev-parse --show-toplevel.

2. README & Commit History

I created a README.md containing my student information, GitHub username, a project description, technologies (HTML, CSS, JavaScript, Git, GitHub and VS Code), a Git command example, a documentation link, and the supplied banner image. I organized commits by logical changes, including initial setup, HTML introduction, CSS styling, learning goals and notes. [VERIFY ACTUAL COMMIT HISTORY AND ATTACH SCREENSHOT.]

3. Selective Staging

I edited index.html and notes.txt before staging. I ran git add index.html to stage only the HTML change and used git status --short to verify that index.html was staged while notes.txt remained modified but unstaged. I committed the HTML change first and later committed notes.txt separately. [ATTACH SELECTIVE-STAGING SCREENSHOT; VERIFY COMPLETION.]

4. Branch & Merge

I created feature/about using git switch -c feature/about, added an About section to index.html and committed that feature on the branch. I switched back to main and merged with git merge --no-ff feature/about. I checked the graph with git log --oneline --graph --all --decorate. [ATTACH BRANCH GRAPH; VERIFY COMPLETION.]

5. Remote Changes

A commit records staged changes in local Git history, while push sends local commits to GitHub. Fetch downloads remote commits and updates remote-tracking references without integrating them into the current branch; pull fetches and integrates changes. [PENDING: Edit README directly on GitHub, run git fetch origin, capture local main behind origin/main, then git pull --ff-only origin main. Capture unstaged and staged diffs and final history.]

6. Fair Use of AI

I used ChatGPT as a learning assistant to understand the Git assignment and receive step-by-step instructions for VS Code. My questions included how to initialize the repository, configure .gitignore, write a README, stage selected files, make commits, create and merge a feature branch, and publish to GitHub. I used the suggested commands as examples and adjusted the repository name to my GitHub username. I checked results using Git

commands such as git status, git log, git branch, and git rev-parse, and reviewed file changes in VS Code. Before submission, I will verify every claim against my own repository and correct any inaccuracies.

7. Sources & Reflection

Sources: Assignment 1 - Git & GitHub Fundamentals (course handout); Git documentation (https://git-scm.com/docs); ChatGPT guidance (used for explanations and example commands).

The most challenging concept was distinguishing the current working directory from the Git repository root and understanding selective staging. If Git behaves unexpectedly, I would first check pwd, git rev-parse --show-toplevel, and git status to confirm the folder, branch, and file states. I would then inspect git log and git remote -v if the issue involved commits or GitHub synchronization.